

CityTech TTP Summer Bootcamp

Intro to React, HTML, and CSS

2026-06-08

Agenda

1. Review
2. Follow Up: Issues and PRs
3. Fishbowl Stand-up
4. HTML
5. React
6. JavaScript and TypeScript
7. Continue Coding

Review

What are the most important things you remember from last week?

Follow Up: Issues and PRs

On the **main course repo**:

- Keep using **GitHub issues** to ask content questions — be **specific**
- Try to figure it out yourself first; use issues as a **secondary** resource
- Someone on the instructional team will answer as soon as we can
- Want to improve the materials? Open a **pull request** — keep PRs **small and specific** in scope

Fishbowl Stand-up

The instructional team will **model a stand-up** while you watch.

- Each person: what I did **yesterday**, what I'm doing **today**, any **blockers**
- Notice how short and focused it stays

Websites and Web Apps

A **website** presents content; a **web app** is interactive software running in the browser. Both are built from the same core technologies:

- **HTML** — the structure and content
- **CSS** — the styling and layout
- **JavaScript** — the behavior and interactivity

HTML and CSS are the foundation — even React ultimately produces HTML and CSS for the browser to render.

HTML — A Markup Language

HTML = HyperText Markup Language.

- A **markup language** wraps content in **tags** that describe its structure and meaning — a heading, a paragraph, a list, an image
- It **describes** content; it doesn't compute (no logic or loops like a programming language)
- The browser reads your HTML and builds the page from it

A First Look at HTML

```
<article>
  <h1>Chocolate Chip Cookies</h1>
  
  <p>Ready in 30 minutes – makes about two dozen.</p>
  <a href="/recipes">More recipes</a>
</article>
```

- Nested tags are **indented** to show the structure
- `<img />` is a **self-closing tag** — no separate closing tag needed
- Tags take **attributes** that configure them — e.g. `<img>`'s `src` (the image file) and `alt` (a text description for screen readers, or if it fails to load)
- Whether an attribute is **required or optional** depends on the tag

Official HTML References

- **MDN Web Docs — HTML** (the go-to reference)
<https://developer.mozilla.org/en-US/docs/Web/HTML>
- **web.dev — Learn HTML** (Google)
<https://web.dev/learn/html>
- **WHATWG HTML Living Standard** (the official spec)
<https://html.spec.whatwg.org/>

Code Together: Dummy HTML Chat App

Let's hand-write a static chat screen in plain **HTML** — no CSS or JavaScript yet.

- A heading with the chat room's name
- A list of messages (hardcoded for now)
- A text input and a **Send** button

"Dummy" = it's just structure; nothing actually sends yet.

Why React Emerged

- Early web pages were **static**; making them interactive meant hand-editing the DOM, step by step
- As **SPAs** grew, keeping the screen in sync with the data by hand became messy and bug-prone
- **Facebook open-sourced React in 2013** to manage complex, constantly-updating UIs (news feed, chat, notifications)
- Today it's one of the most widely used ways to build web interfaces

What Is React?

- A **JavaScript library** for building user interfaces
- **Components** — reusable, self-contained pieces of UI
- **Declarative** — you describe what the UI should look like; React updates the page
- **JSX** — write HTML-like markup right inside JavaScript

What Is a Function?

In programming, a **function** takes some **input**, does work, and returns an **output**.

- Same input → same output — predictable
- **Reusable** — call it again with different inputs instead of repeating yourself

```
function greet(name) {  
  return "Hello, " + name;  
}  
greet("Maya"); // "Hello, Maya"
```

The Big Idea: $UI = f(state)$

A component is a **function of its data** — give it data, it returns UI. Change the data and React re-renders just that component.

```
function MessageList({ messages }) {  
  return (  
    <ul>  
      {messages.map((m) => <li key={m.id}>{m.text}</li>)}  
    </ul>  
  );  
}
```

- **Static HTML:** you'd hand-write and edit every `<li>` yourself
- **A function:** the list is generated from `messages` — no duplication

React References

- **React docs (home)** — <https://react.dev>
- **Learn React** (guided tutorial) — <https://react.dev/learn>
- **API reference** — <https://react.dev/reference/react>

Real-World React Frontends

Open-source React apps worth exploring:

- **Excalidraw** — virtual whiteboard, client-side React + TypeScript
<https://github.com/excalidraw/excalidraw>
- **tldraw** — infinite canvas and drawing
<https://github.com/tldraw/tldraw>
- **TodoMVC** — the classic to-do app, small and readable
<https://github.com/tastejs/todomvc>

What Is Boilerplate?

Boilerplate is the standard, repeated setup code and config every project needs *before* you write any features — folder structure, build config, linting, base dependencies.

- The same scaffolding shows up across countless projects
- Teams start from **starter templates / generators** instead of reinventing it
- Saves time and enforces consistency across a codebase
- Trade-off: it can include things you don't need — know what it sets up

Start a React + TypeScript Project

We use **Vite** (modern and fast) with the React + TypeScript template — and **Yarn**, not npm.

```
yarn create vite my-app --template react-ts
cd my-app
yarn          # install dependencies
yarn dev      # start the dev server
```

- `react-ts` gives you React + TypeScript boilerplate, ready to go
- Open the printed `localhost` URL to see it live

Build Tools and Dev Servers

Two pieces of modern web tooling:

- **Build tool** — turns your source (many TS / JSX / CSS files) into optimized files a browser can run (`src/` → `dist/`)
 - bundled, minified, **tree-shaken** (unused code dropped), and obfuscated
- **Dev server** — runs your app locally while you develop, with **hot reload**: save a file and the change appears instantly — no manual refresh or rebuild

What Is Vite?

Vite is a modern tool that is **both** a build tool and a dev server.

- `yarn create vite` — scaffolds the project (your boilerplate)
- `yarn dev` — starts the dev server with hot reload
- `yarn build` — produces the production build (`src/` → `dist/`)

`src/` vs. `dist/`

- `src/` — your source code: the files you write (TypeScript, JSX, CSS)
- `dist/` — the build output: optimized, browser-ready files generated from `src/`
- You **edit** `src/`; you **never hand-edit** `dist/` — it's regenerated each build
- The browser ultimately runs what's in `dist/`

Compilation and Building

Compilation turns your source into what browsers can actually run: TypeScript → JavaScript, JSX → JS — then it's bundled and minified.

```
yarn build          # compile src/ → dist/ once  
yarn build --watch  # rebuild automatically on every save
```

- `yarn build` runs the project's build script (Vite) to produce `dist/`
- `--watch` keeps watching your files and rebuilds as you change them

What Is Yarn? (vs. npm)

Both are **package managers** — they install and manage the libraries your project depends on (everything in `node_modules`).

- **npm** ships with Node; **Yarn** is a popular alternative — the one we use
- Both read `package.json` and write a lockfile for reproducible installs (`yarn.lock` vs. `package-lock.json`)

```
npm install           → yarn
npm install <pkg>     → yarn add <pkg>
npm run <script>      → yarn <script>
npm uninstall <pkg>  → yarn remove <pkg>
```

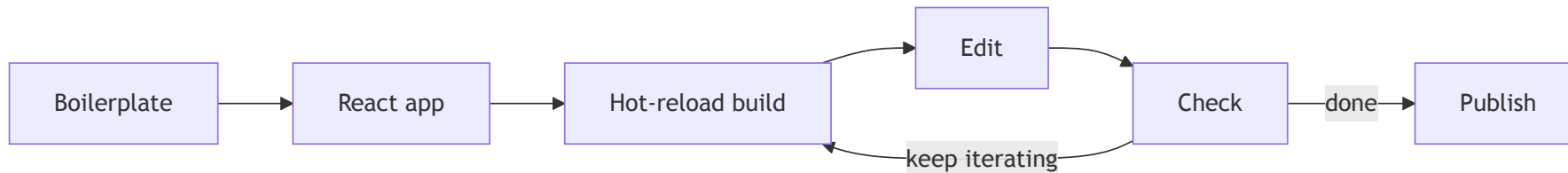
Why SPAs? (Reminder)

Traditional sites reload the **whole page** from the server on every click. A **single-page app (SPA)** loads once, then updates only the parts that change.

- **App-like feel** — no jarring full-page reloads
- **Less data** after the first load — fetch just the data, not whole pages
- **UI keeps its state** as you move around
- React is built for this: components re-render when their data changes

Pause and Check Your Mental Model

The development loop, start to finish:

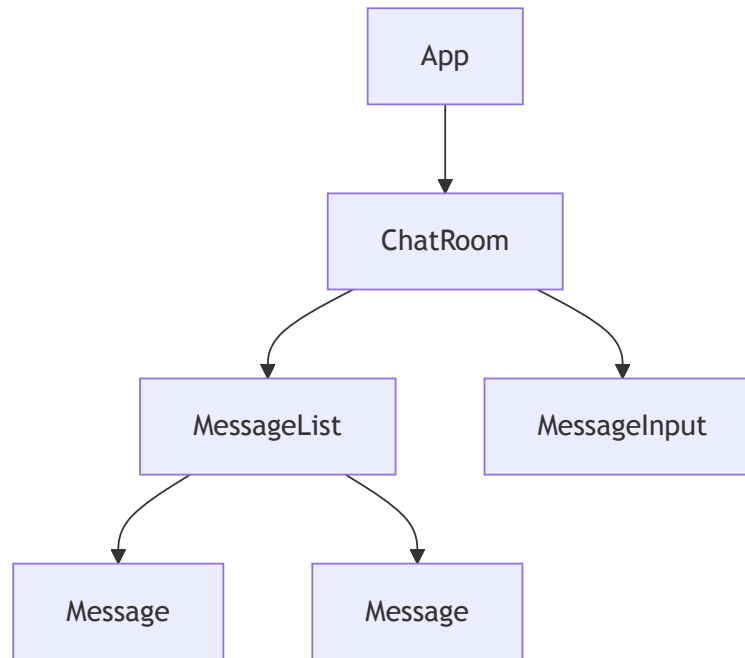


Components: The Mental Model

- A **component** is a function: **props in** → **UI (JSX) out**
- You build your interface as a **tree** of nested components
- They're **reusable building blocks** — think LEGO bricks
- A component can hold its own **state** and re-renders when that state changes

Composing Components

Break a screen into components, then nest them into a tree. **Props flow down** (parent → child); **events flow up** (child → parent via callbacks).



Code Together — Build a Fake Chatroom

Let's build a simple chat UI together — no real backend, messages just live in React **state**.

- **ChatRoom** — holds the list of messages in state
- **MessageList** — renders all the messages
- **Message** — displays a single message
- **MessageInput** — text box + send button that adds a message

"Fake" = messages are stored in state, not sent anywhere.

Intro to JavaScript

JavaScript (JS) is the programming language of the web — it runs in every browser and adds **behavior** to a page.

- Respond to clicks, update content, fetch data
- Runs in the browser (and on servers via Node)

JavaScript Everywhere

Thanks to **Node.js** — a runtime that runs JavaScript outside the browser — one language spans the whole stack:

- **Frontend** — in the browser (React, etc.)
- **Backend** — servers and APIs (Node.js, Express)
- **Desktop** — Electron (VS Code, Slack)
- **Mobile** — React Native

This is what makes **full-stack development** in a single language possible. (Code that runs on both client and server is called **isomorphic / universal JavaScript**.)

Intro to TypeScript

TypeScript (TS) is JavaScript **plus types** — it's built **on top of** JavaScript and compiles down to plain JS.

- JavaScript is **dynamically typed** — types are only checked as it runs
- TypeScript adds **static types** — mistakes are caught *before* it runs
- **Any valid JavaScript is valid TypeScript** — use pure JS, or mix JS and TS freely
- The browser never runs TS directly; it's compiled to JavaScript first

add — JavaScript vs. TypeScript

```
// JavaScript
function add(a, b) {
  return a + b;
}
```

```
// TypeScript
function add(a: number, b: number): number {
  return a + b;
}
```

- **Dynamic typing (JS)** — `a` and `b` can be anything; types are checked only when it runs, so a bad type fails at runtime
- **Static typing (TS)** — `a` and `b` are declared as `number`; a wrong type is caught as *you write*, before it runs

JavaScript and TypeScript References

- **MDN — JavaScript** (the JS reference)
<https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- **javascript.info** (modern JS tutorial)
<https://javascript.info/>
- **TypeScript Handbook** (official docs)
<https://www.typescriptlang.org/docs/>

Continue Coding

Keep working on your dummy chat app.

- What else do you think it needs?
- Can you think of a way to **simulate a person chatting back**?
- Capture your thoughts in a **design doc** — or at least a **TODO.md** file

Design Docs and TODO Files

- A **design doc** is a short written plan — *what* you're building, *why*, and *how* (approach, structure, open questions) — written as you design, before you write much code
- A **TODO.md** is a simple running checklist of tasks and ideas, kept in the repo
- Both make your thinking visible, keep you organized, and look great on your GitHub
- On a **team**, they're how everyone stays **in sync** — a shared plan and a shared to-do list